

MACHINE LEARNING PROJECT

Product Purchase Prediction Based on Ads

Social Media Ad Analytics • Random Forest Classification

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AIM

To predict whether a customer will purchase a product based on social media advertisement data using machine learning classification techniques.

Leverage demographic & behavioral data from ad interactions

Provide marketers with actionable purchase intent signals

Improve ad targeting ROI through predictive analytics

OBJECTIVES

01 To analyze customer behavior based on social media advertisement data

02 To preprocess and clean the dataset for effective modeling

03 To select important features influencing purchase decisions

04 To build a machine learning model using Random Forest

05 To predict whether a customer will buy a product or not

06 To provide probability of purchase for better decision making

07 To suggest suitable products based on predicted customer interest

08 To develop a user-friendly web interface for real-time prediction

DATASET & KEY FEATURES

Feature	Type	Description
User ID	Categorical	Unique identifier for each social media user
Gender	Binary	Male / Female classification
Age	Numerical	User's age — key purchase predictor
Estimated Salary	Numerical	Annual income proxy
Purchased	Target (0/1)	Whether user bought the advertised product

Source: Social Network Ads Dataset | Downloaded from Kaggle

PROCEDURE

Step 1: Data Collection

Loaded Social Network Ads CSV dataset into VS Code

Step 2: Preprocessing

Dropped irrelevant columns, handle nulls, normalize age & salary

Step 3: Feature Selection

Selected Age & Estimated Salary as input features; Purchased as label

Step 4: Model Training

Splitted 80/20 train-test, fit classifier with cross-validation

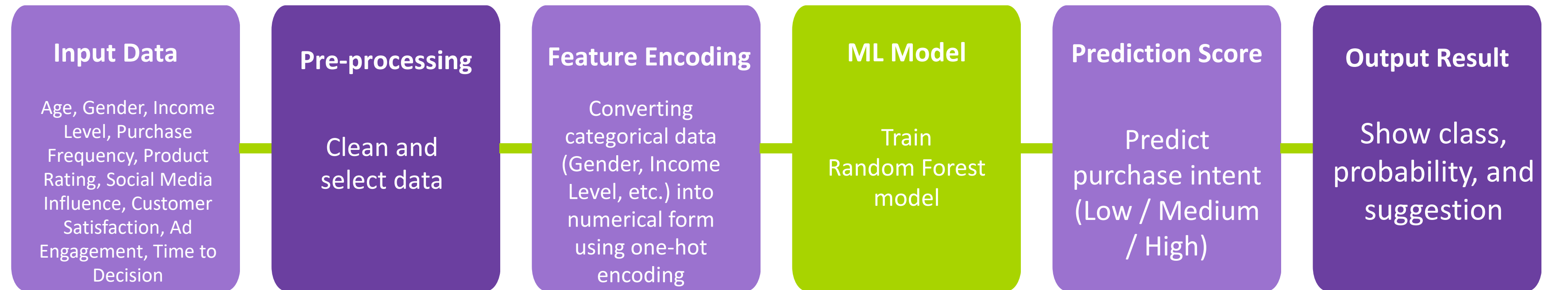
Step 5: Evaluation

Confusion matrix, accuracy, ROC-AUC score aren't used for evaluation

Step 6: Deployment

Built Streamlit/Tkinter UI for real-time user predictions

MODEL WORKFLOW



Technology Stack

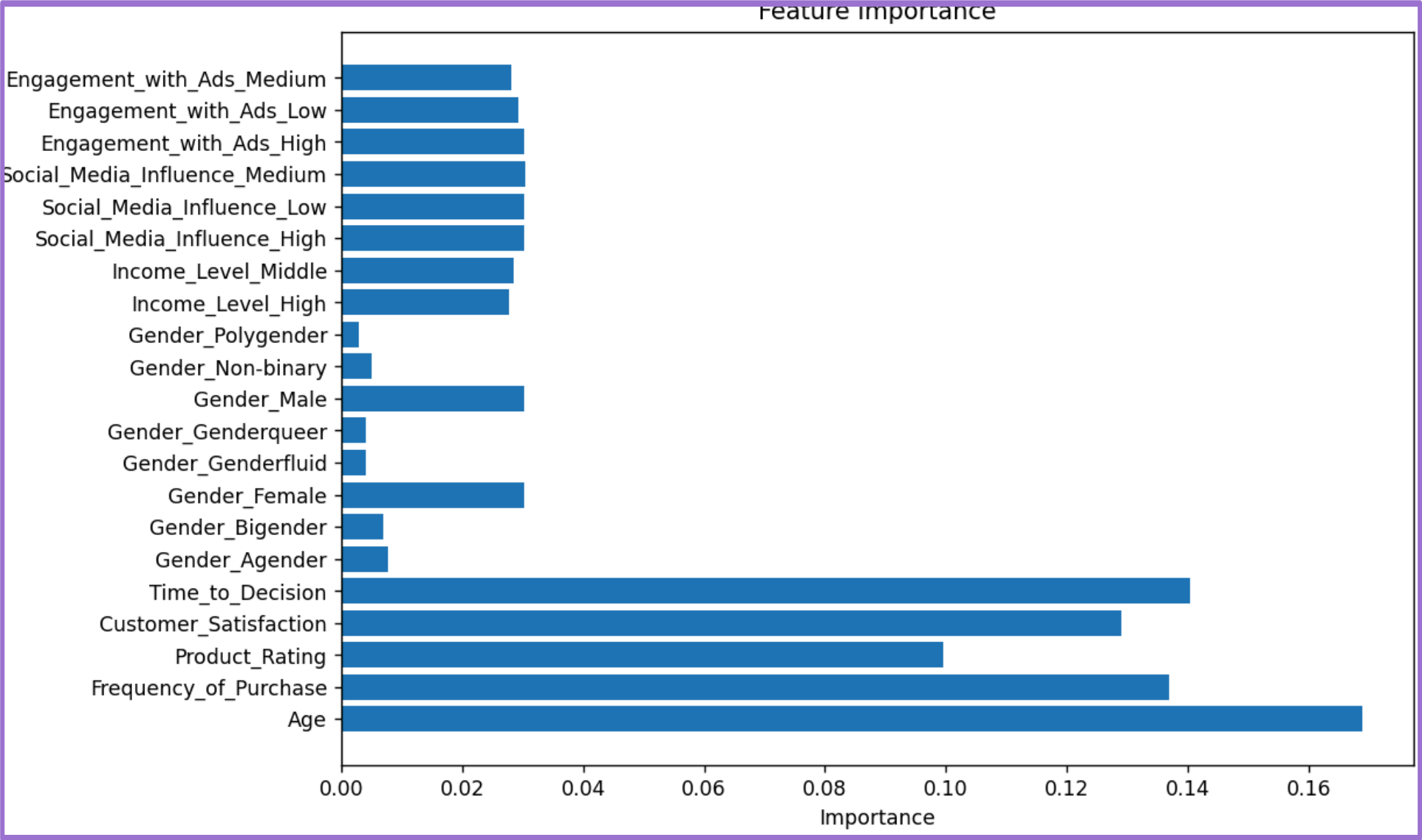


GRAPHS & VISUALIZATIONS

Bar chart showing Age vs. Salary impact

Feature Importance Chart

- Income Level and Purchase Frequency have strong influence on prediction
- Product Rating and Customer Satisfaction also play an important role
- Social Media Influence and Ad Engagement affect buying behavior
- Age has moderate impact on decisions
- Gender has very low impact and is mostly ignored by the model



GRAPHS & VISUALIZATIONS

CLASS DISTRIBUTION GRAPH

CLASS DISTRIBUTION GRAPH

Balanced dataset:

- All four classes (Need-based, Wants-based, Planned, Impulsive) have almost equal counts, meaning the data is well balanced.

No class dominance:

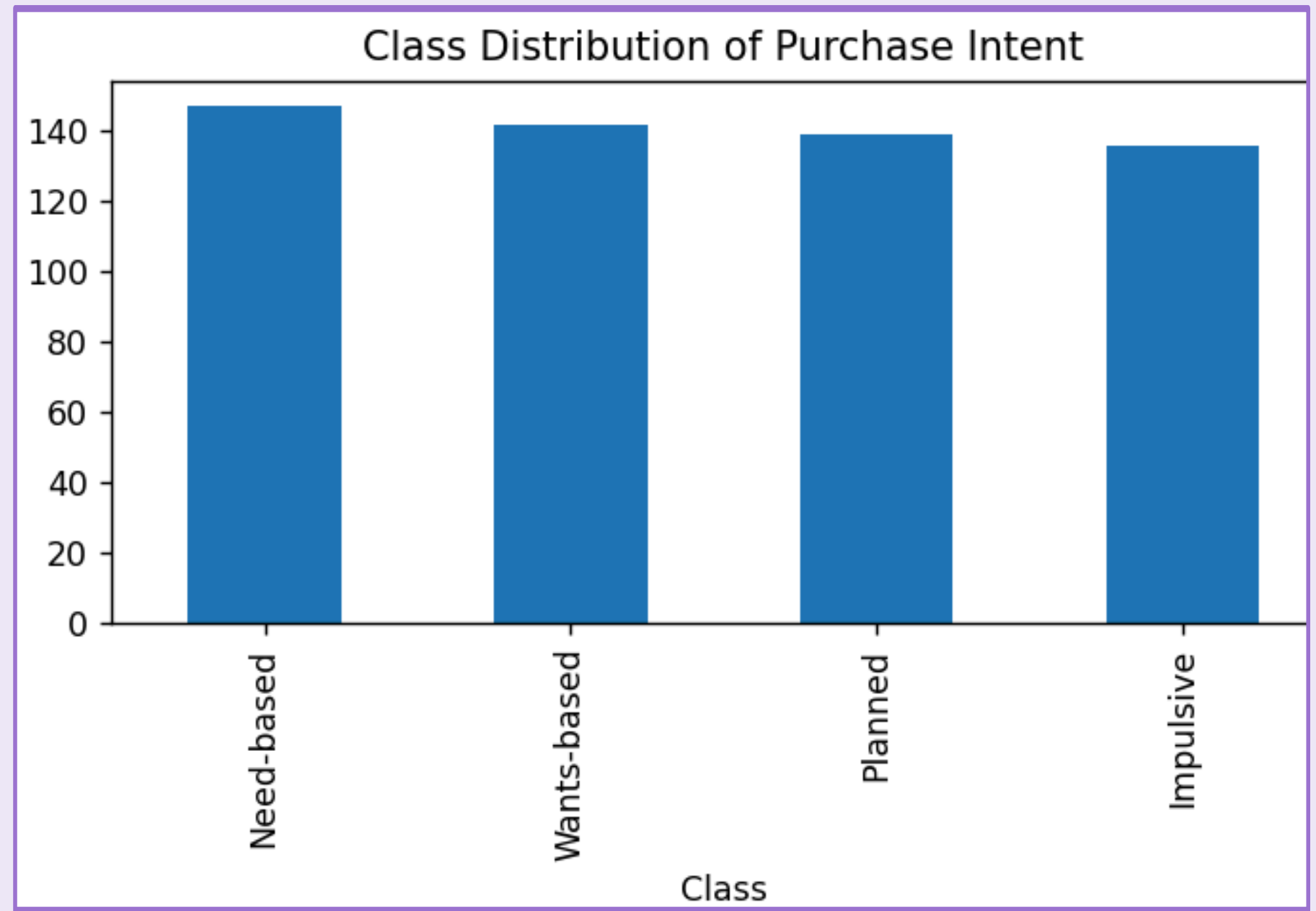
- No single category is significantly higher than others, so the model is not biased toward one class.

Good for training:

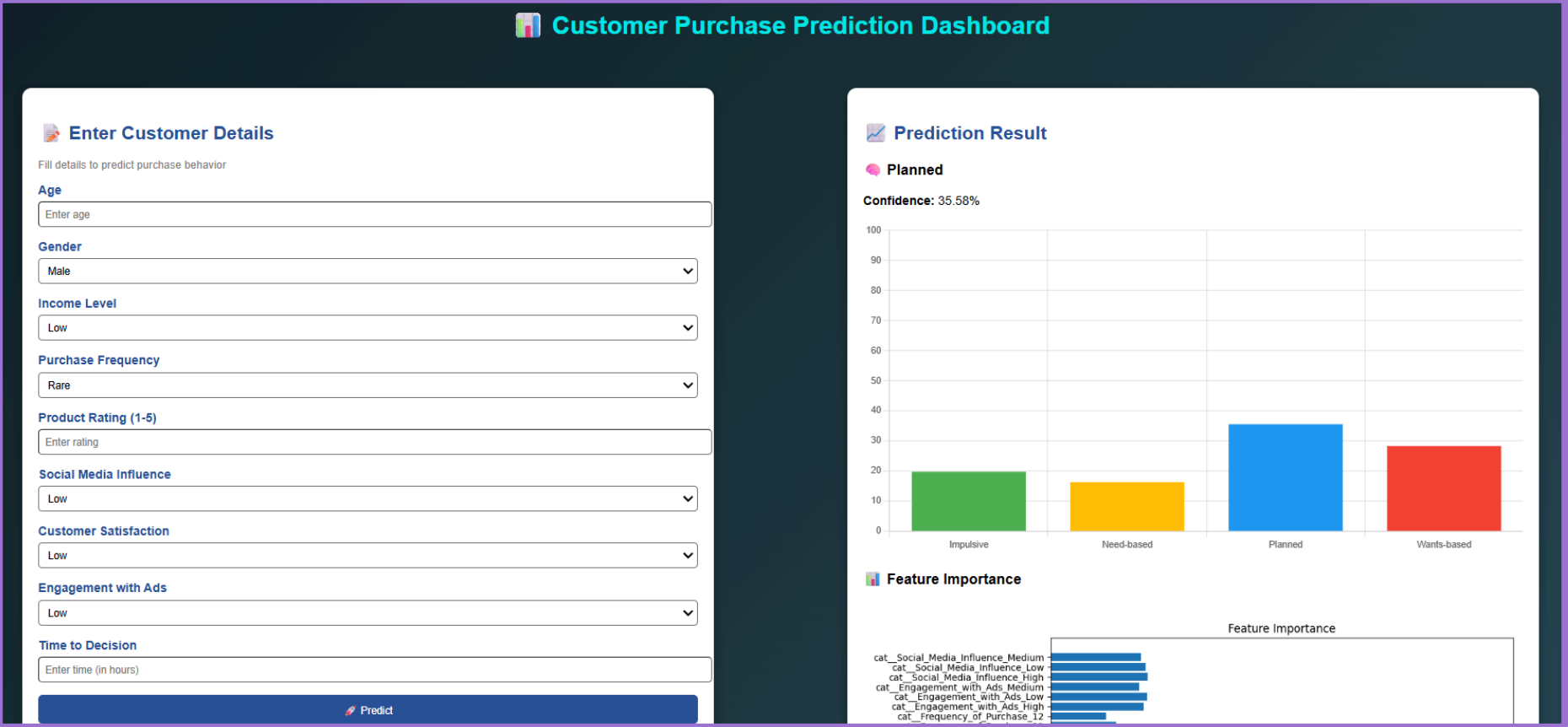
- Balanced data helps the model learn all classes properly and improves fairness in prediction.

Slight variation exists:

- Small differences in counts are present, but they are minimal and won't strongly affect model performance.



USER INTERFACE



Input Form

Fields for Age, Gender, Estimated Salary

Prediction Result

Shows probability score + class of the custoer predicted

Visual Output

Gauge chart of purchase likelihood

RESULTS & CONCLUSION

- Model shows moderate performance due to multi-class classification
- Customer behavior features influence predictions more than demographic features
- Model successfully provides purchase prediction with probability
- Product suggestions help in targeted marketing decisions